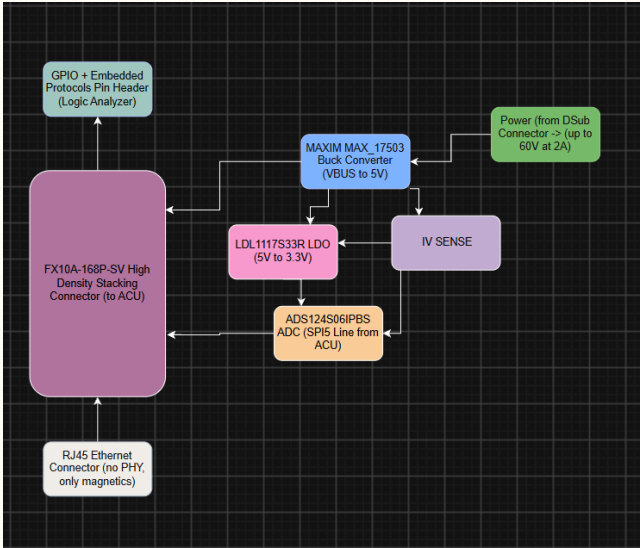


- Things to do (Bhuvi) - 9/6:
- coplanar gnd pour on signal 1
 - silk screen (pinout)
 - serial number
 - ethernet tvs diode ground via



ACU IO Board

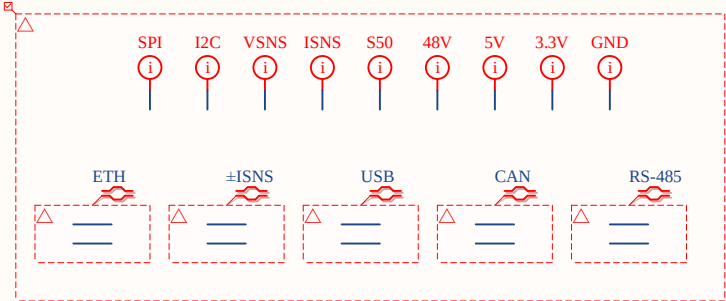
Responsible Engineer: Bhuvanesh Senthil



- Informational Notes
- Layout Notes
- Assembly Notes
- Software Notes
- Todo Items

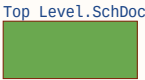
Some of these directives have predefined rules in the PCB

Feel free to ignore these

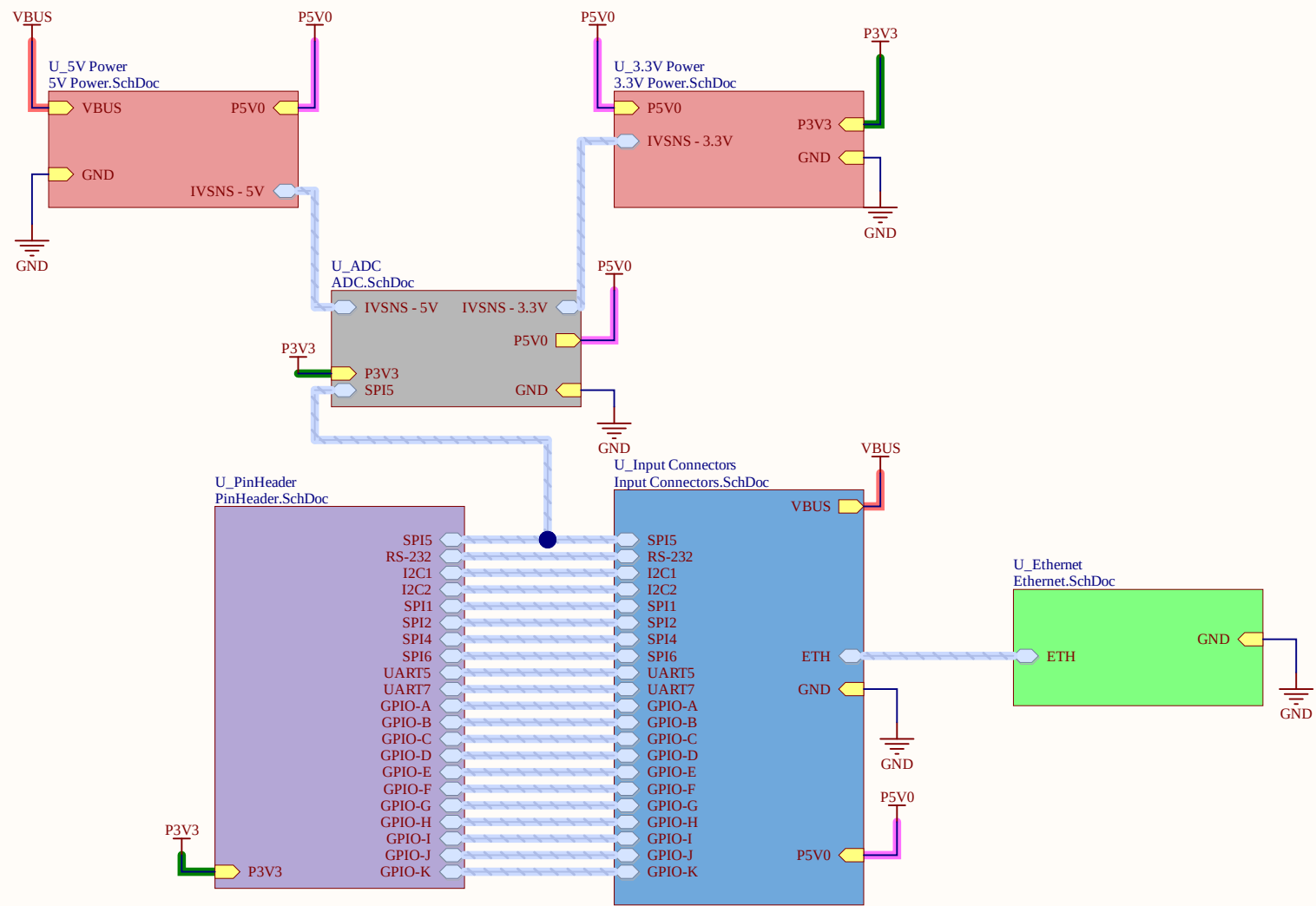


Project: ACU IO Board.PrjPcb				
Sheet Title: Cover Page.SchDoc				
Size: A4	Schematic Designer: Bhuvanesh Senthil PCB Layout Designer: Bhuvanesh Senthil Responsible Engineer: Bhuvanesh Senthil			Yellow Jacket Space Program Georgia Tech Atlanta, GA
Date: 9/8/2026	Revision: 1.0	Sheet: 1	of: 8	

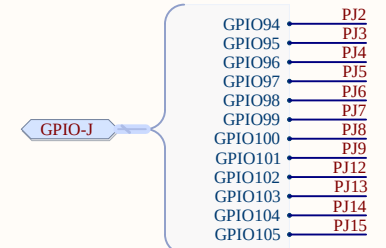
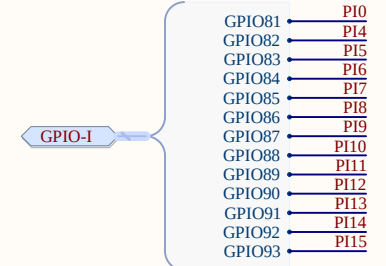
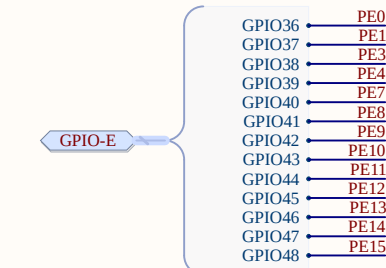
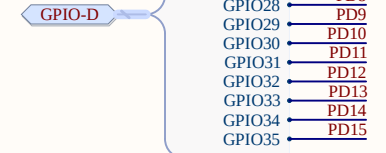
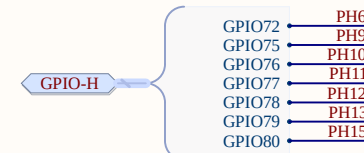
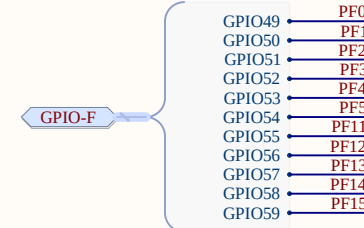
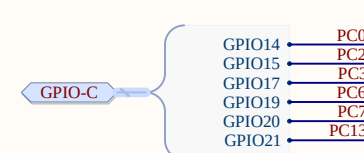
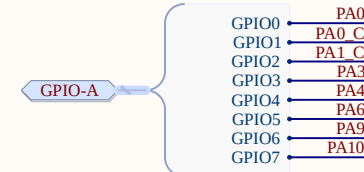
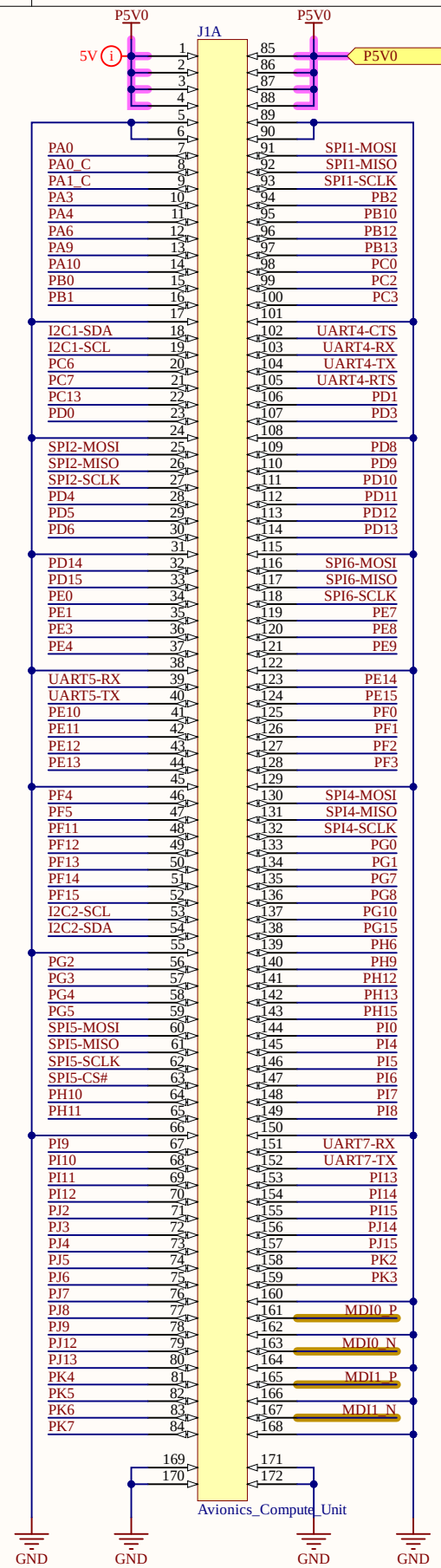
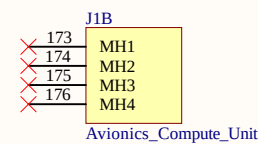
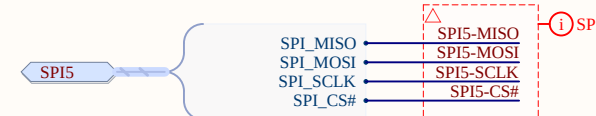
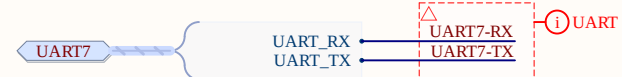
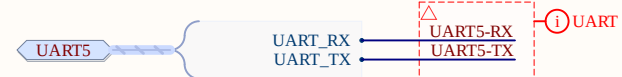
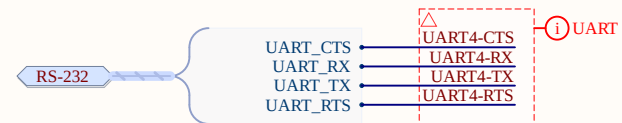
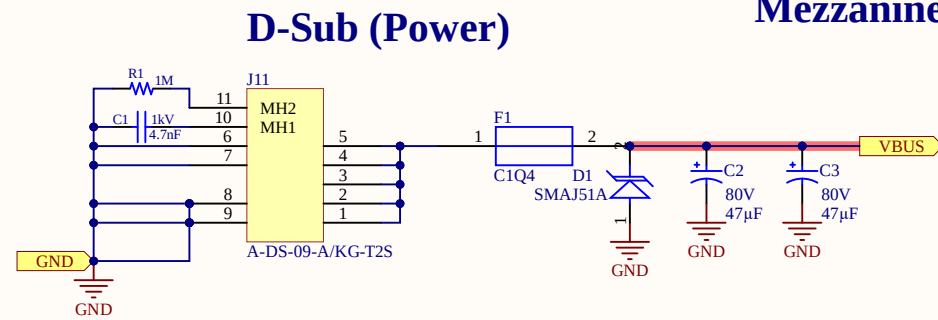
- MH1
- M3_PTH
- MH2
- M3_PTH
- MH3
- M3_PTH
- MH4
- M3_PTH



Top Level Diagram



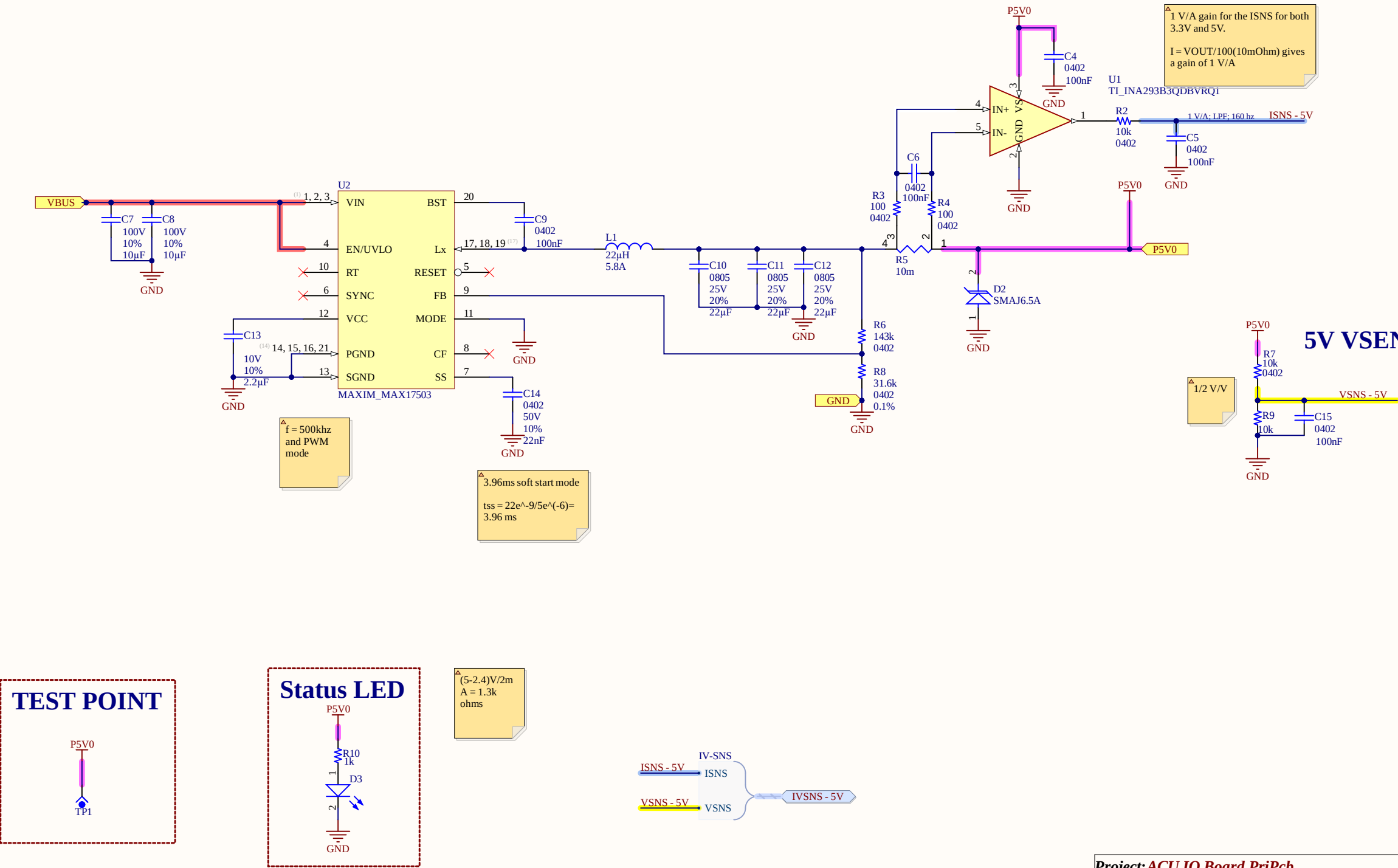
Input Connectors



5V Buck

5V ISENSE

5V VSENSE



Project: ACU IO Board.PrjPcb			
Sheet Title: 5V Power.SchDoc			
Size: A4	Schematic Designer: Bhuvanesh Senthil	Yellow Jacket Space Program Georgia Tech Atlanta, GA	
	PCB Layout Designer: Bhuvanesh Senthil		
	Responsible Engineer: Bhuvanesh Senthil		
Date: 9/8/2026	Revision: 1.0	Sheet: 4	of: 8



3.3V LDO

3.3V ISENSE

Δ 1 V/A gain for the ISNS for both 3.3V and 5V.
I = VOUT/100(10mOhm) gives a gain of 1 V/A

3.3V VSENSE

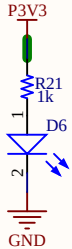
Δ 1/2 V/V

Δ (3.3-2.4)/2m
A 450 ohms

TEST POINT



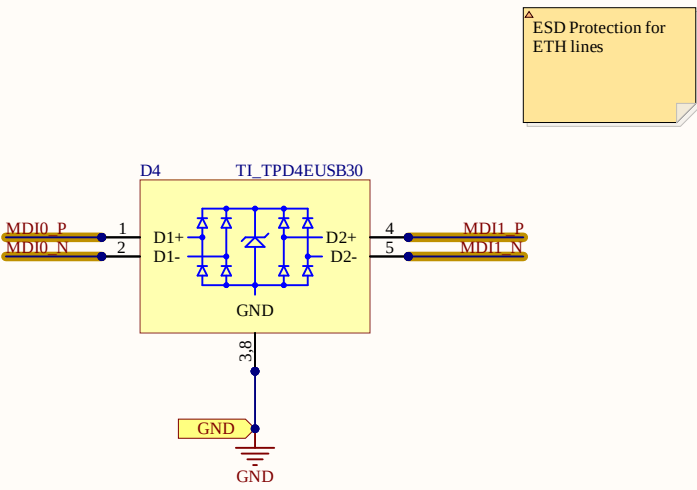
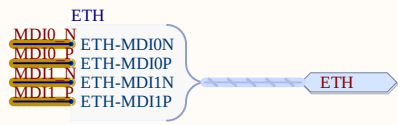
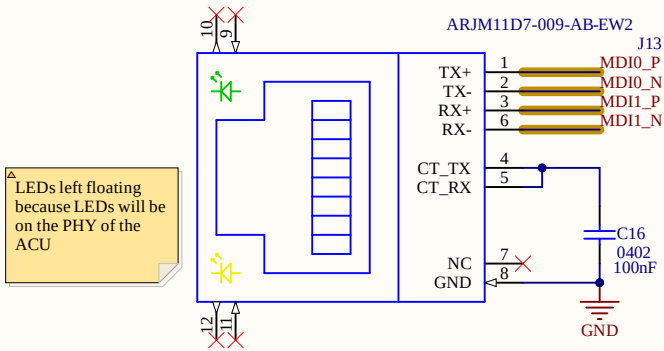
Status LED



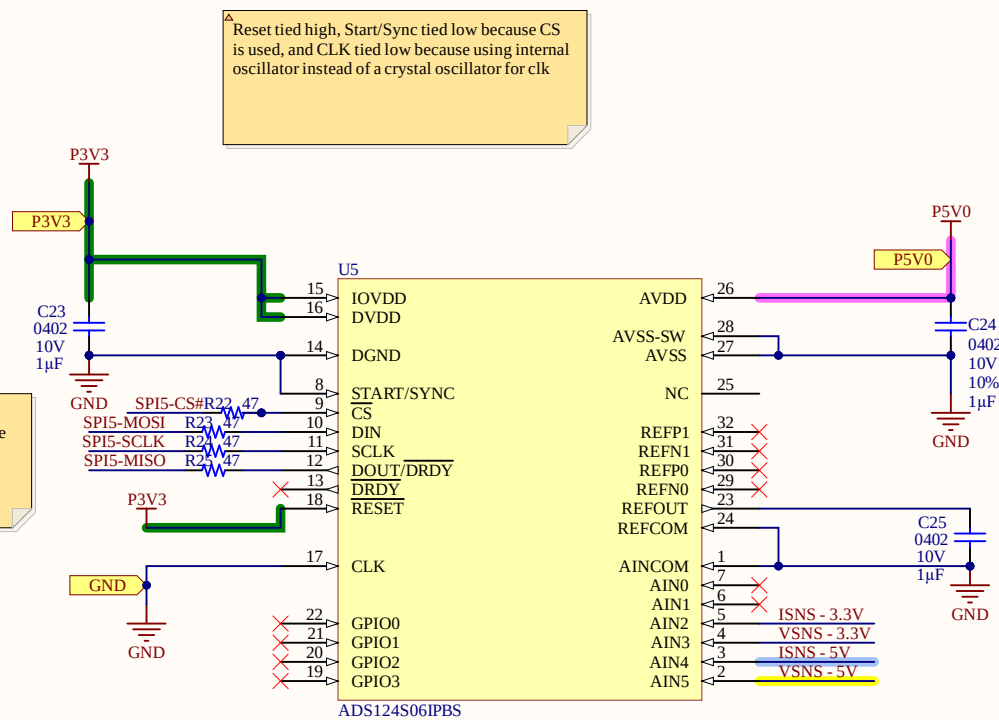
Project: ACU IO Board.PrjPcb				
Sheet Title: 3.3V Power.SchDoc				
Size: A4	Schematic Designer: Bhuvanesh Senthil PCB Layout Designer: Bhuvanesh Senthil Responsible Engineer: Bhuvanesh Senthil			<i>Yellow Jacket Space Program</i> <i>Georgia Tech</i> <i>Atlanta, GA</i>
Date: 9/8/2026	Revision: 1.0	Sheet: 5	of: 8	



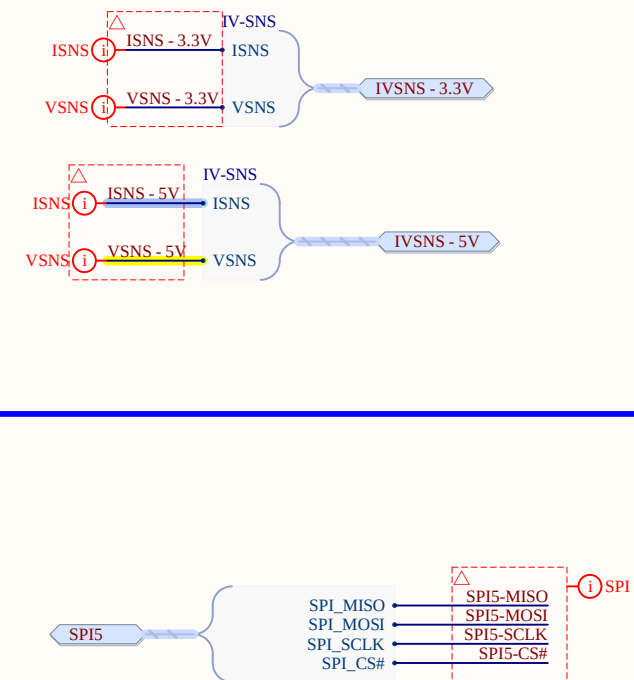
Ethernet



ADC



Power IVSENSE

Project: **ACU IO Board.PrjPcb**

Sheet Title: **ADC.SchDoc**

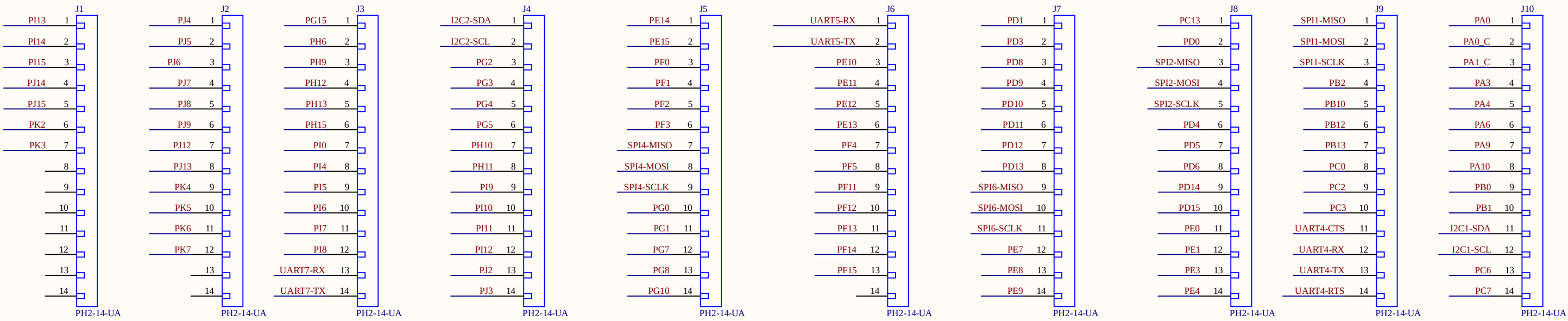
Size: A4	Schematic Designer: Bhuvanesh Senthil PCB Layout Designer: Bhuvanesh Senthil Responsible Engineer: Bhuvanesh Senthil
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Yellow Jacket Space Program
Georgia Tech
Atlanta, GA

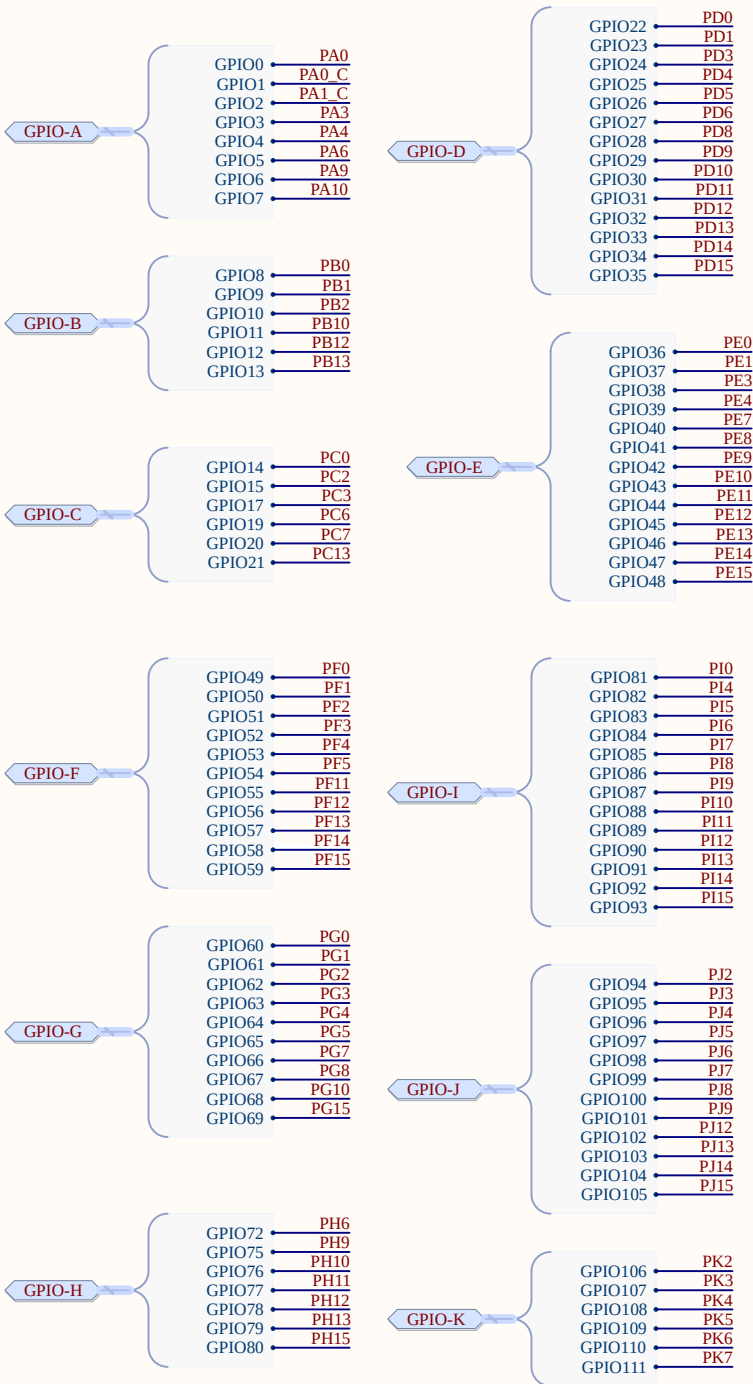


Output Connectors

Pin Headers



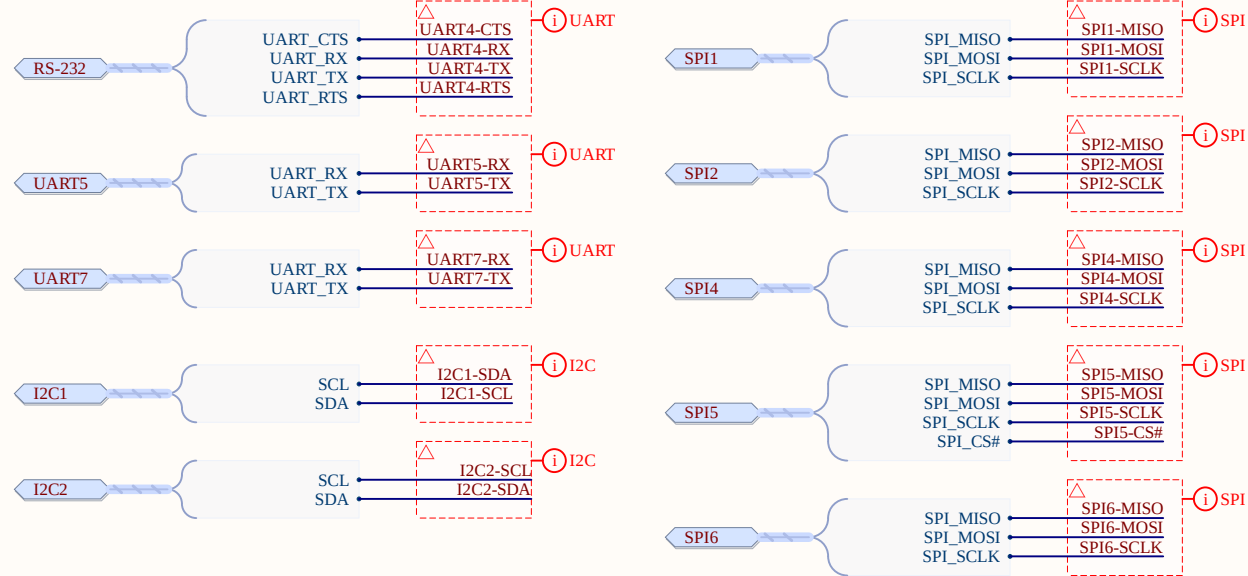
GPIO Bank



I2C Pullup Resistors



Protocol Bank



Protocols and GPIO on Pin Header to be tested via a logic analyzer

SPI5 protocol used on ADC and tested via ADC for IV SENSE